

TravelCrew-Multi-Agent-Hopitality-Creator

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Project Number: : AAI-08

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Academic Year: 2025 – 2026

Problem Statement & Objectives

1. Problem Statement
2. Project Objectives
3. Scope of the Project

Proposed Solution

1. Key features *(Just mention key features here no need to go into details)*
2. Overall Architecture / Workflow
3. Tools & Technologies Used *(If applicable)*

Results & Output

Add the below details here:

1. Screenshots / outputs
2. Reports / dashboards / models
3. Key outcomes

Conclusion

Mention a brief conclusion about your project summing up everything you have worked on and the key learning.

Future Scope & Enhancements

Problem Statement & Objectives

1. Problem Statement

Travel planning is inherently complex and time-consuming, requiring users to gather information from multiple disconnected platforms such as hotel booking sites, travel blogs, maps, and budgeting tools. This fragmented approach leads to inefficiencies, inconsistent information, and increased cognitive load for users.

Additionally, most existing solutions lack personalization and intelligence, forcing users to manually curate itineraries based on scattered data. There is no unified system that can understand user preferences—such as destination, duration, budget, interests, and travel style—and automatically generate a **comprehensive, structured, and optimized travel plan**.

Furthermore, traditional systems do not effectively leverage real-time data or contextual knowledge, resulting in outdated or generic recommendations. This creates a need for an intelligent, automated solution that integrates multiple data sources and delivers **personalized, accurate, and end-to-end travel itineraries**.

2. Project Objectives

The primary objectives of the TravelCrew system are as follows:

- **Develop an AI-powered travel planning system**
Design and implement an intelligent platform capable of transforming user inputs into structured travel itineraries using advanced language models.
- **Implement a multi-agent architecture**
Utilize specialized AI agents (Researcher and Writer) to divide responsibilities, enabling efficient task execution, modularity, and scalability.
- **Integrate Retrieval-Augmented Generation (RAG)**
Combine a vector-based knowledge base (ChromaDB) with large language models to ensure responses are contextually accurate, relevant, and grounded in reliable data.
- **Enable personalized itinerary generation**
Generate customized travel plans based on user-defined parameters such as destination, duration, budget, travel type, and interests.
- **Incorporate real-time data enrichment**
Enhance itinerary accuracy and relevance by integrating live web search (Serper API) to fetch up-to-date information on hotels, attractions, and travel conditions.

- **Ensure modular and scalable system design**

Build the system with a layered architecture that supports easy integration of additional agents, tools, and features in future iterations.

3. Scope of the Project

The TravelCrew system is designed to provide a comprehensive and intelligent solution for automated travel planning.

The scope of the project includes the following key aspects:

- **End-to-End Travel Planning**

The system covers the complete travel planning lifecycle, including recommendations for hotels, restaurants, tourist attractions, transportation options, and budget estimation. It generates a fully structured, day-by-day itinerary, eliminating the need for users to consult multiple sources.

- **Multi-Destination Support**

The system is capable of handling multiple global destinations such as Tokyo, Paris, Bali, and India. It leverages a pre-built knowledge base along with live web data to ensure relevant and location-specific recommendations.

- **AI-Based Multi-Agent Collaboration**

The project utilizes a multi-agent architecture where specialized agents collaborate to perform distinct tasks. The Researcher Agent focuses on data collection and analysis, while the Writer Agent synthesizes this information into a coherent and user-friendly travel itinerary.

- **Integration of Knowledge Base and Real-Time Data**

The system combines a Retrieval-Augmented Generation (RAG) approach using ChromaDB with real-time web search integration. This ensures both accuracy (from stored knowledge) and freshness (from live data).

- **User-Centric Personalization**

The system processes user inputs such as destination, duration, budget, travel type, and interests to generate personalized travel plans tailored to individual preferences.

- **Scalability and Future Extensibility**

The architecture is modular and designed for future enhancements. Additional agents such as Budget Optimization Agent, Weather Agent, or Visa Assistance Agent can be integrated without major system redesign.

- **Web-Based Accessibility**

The application is deployed as a web interface using Streamlit, making it easily accessible and user-friendly without requiring technical expertise.

Proposed Solution

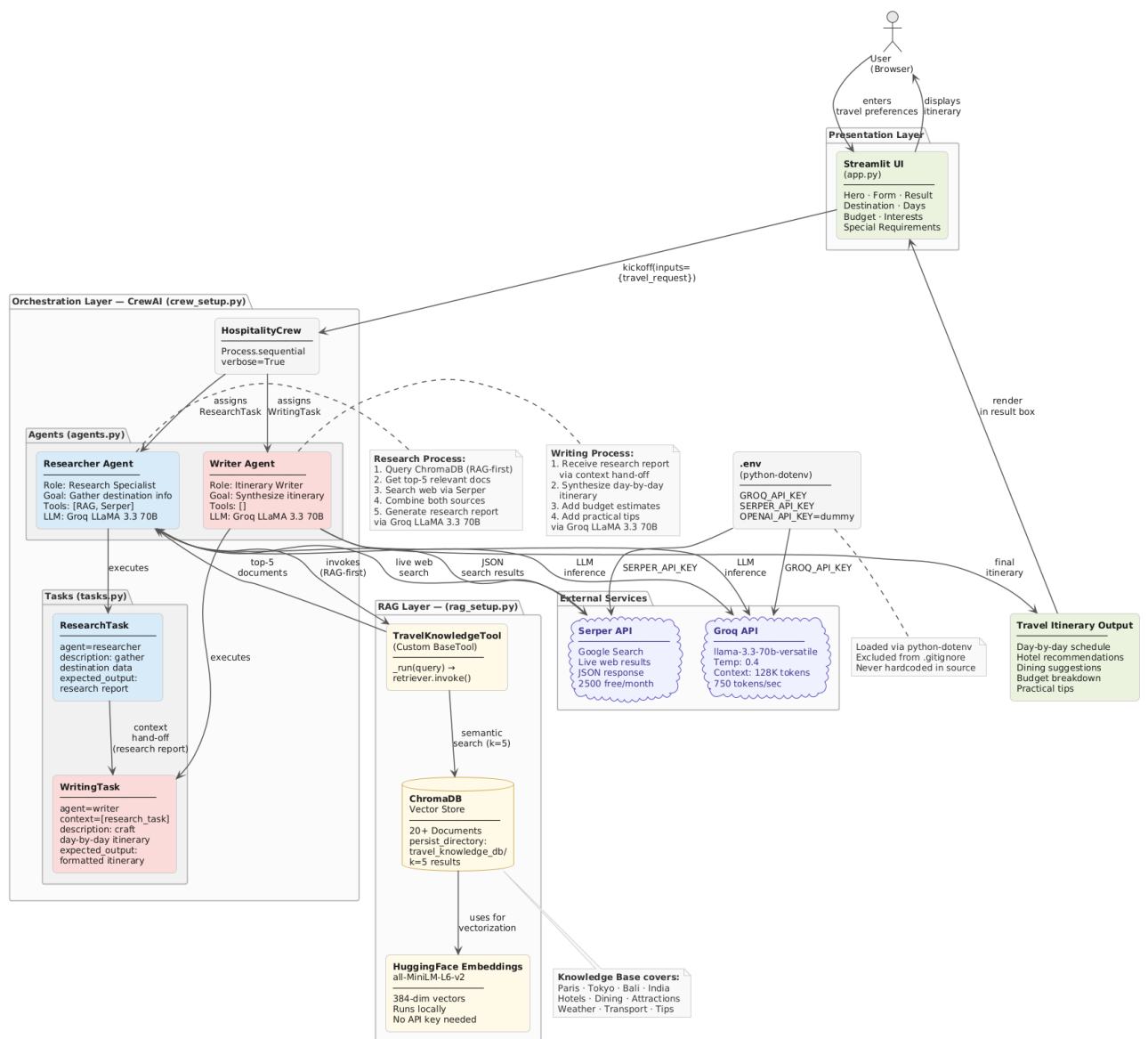
1. Key Features

- Multi-Agent System (Researcher Agent and Writer Agent)
 - RAG-based knowledge retrieval using ChromaDB
 - Real-time web search integration using Serper API
 - LLM-powered itinerary generation using Groq LLaMA 3.3 70B
 - Structured day-by-day travel itinerary
 - Budget estimation with travel recommendations
 - Interactive and user-friendly Streamlit interface
-

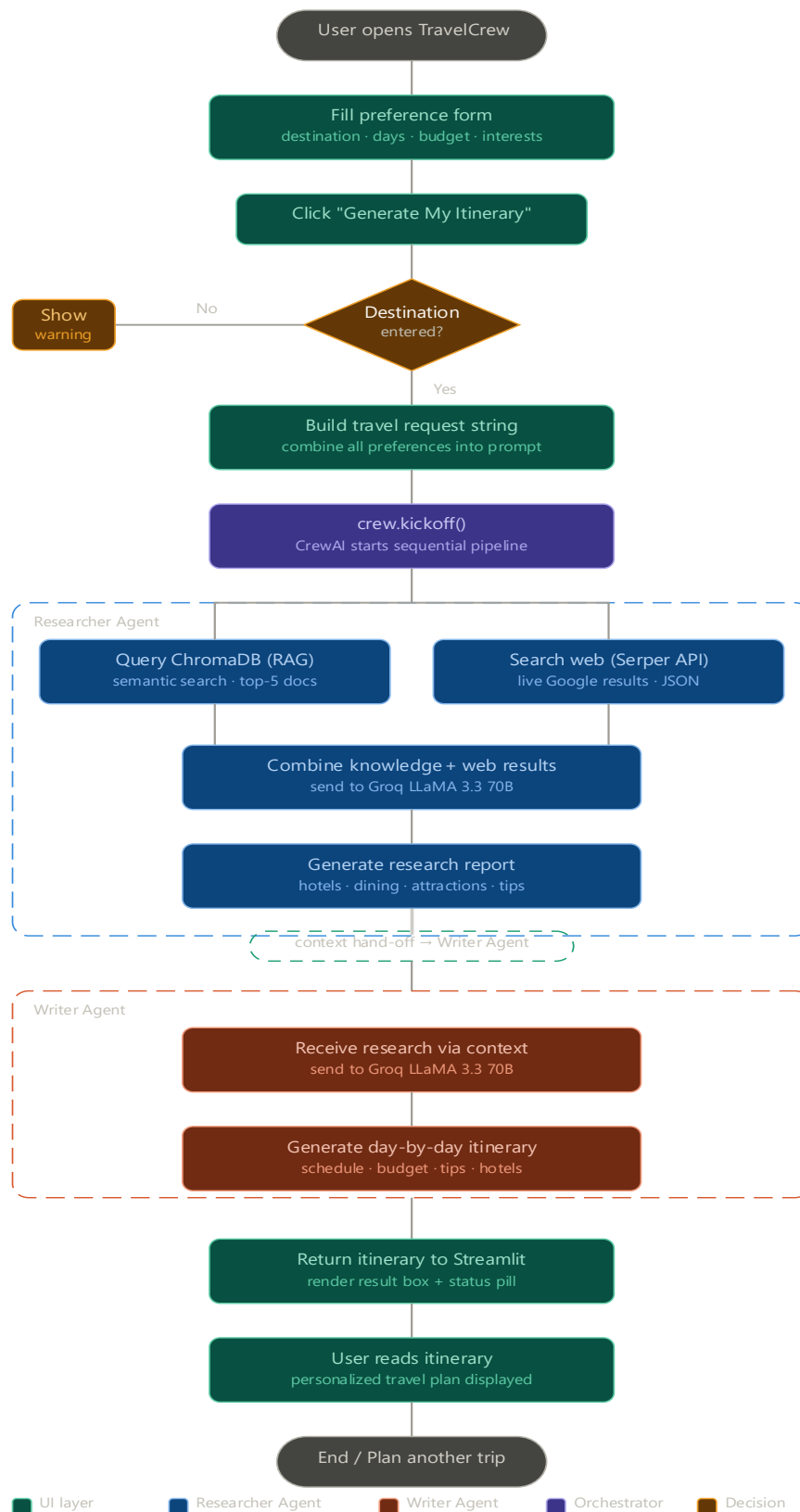
2. Overall Architecture / Workflow

1. Overall Architecture

TravelCrew — Multi-Agent Hospitality System Architecture



2. Process Flow



The system follows a modular, layered architecture with a sequential multi-agent workflow:

1. User inputs travel preferences through the Streamlit UI
2. CrewAI orchestrates the workflow
3. Researcher Agent retrieves data using:
 - ChromaDB (RAG-based knowledge base)
 - Serper API (live web search)
4. Research output is passed via context hand-off
5. Writer Agent generates a structured itinerary
6. Final output is displayed to the user

Information Flow:

User → Streamlit → CrewAI → Researcher Agent → Writer Agent → Output

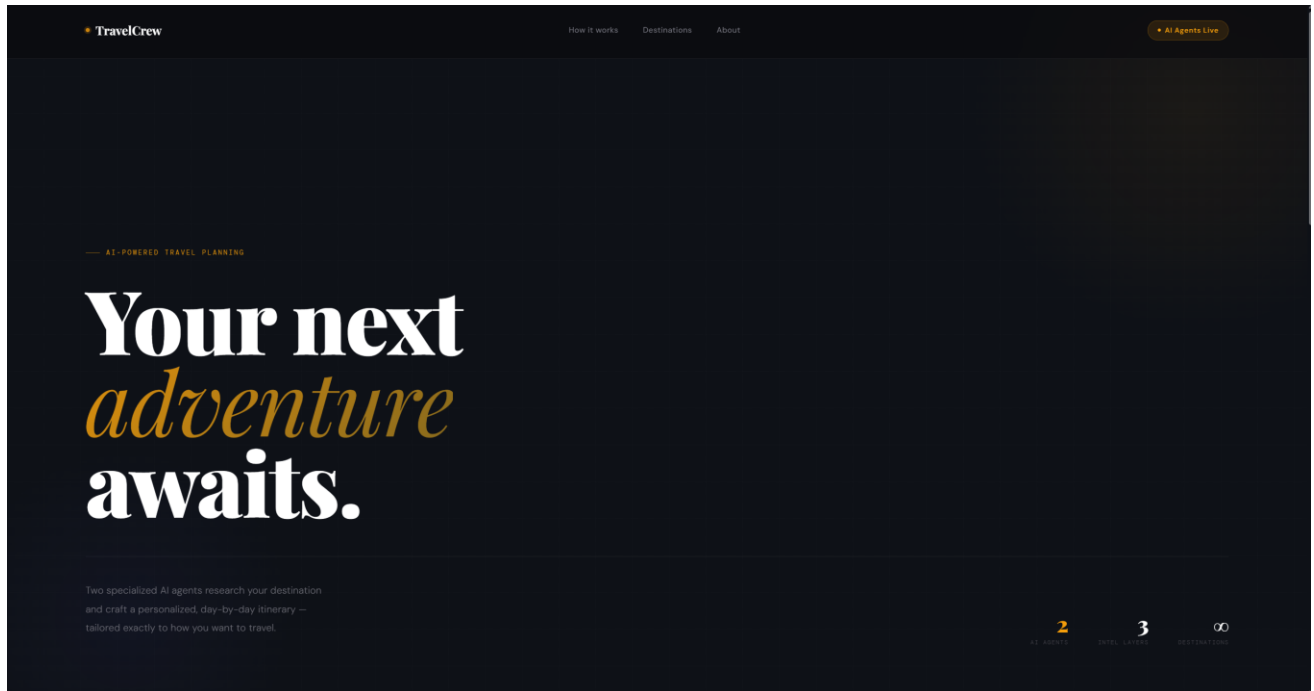
3. Tools & Technologies Used

Category	Technology
Frontend	Streamlit
Backend	Python
AI Framework	CrewAI
LLM	Groq LLaMA 3.3 70B
Vector DB	ChromaDB
Embeddings	HuggingFace (MiniLM-L6-v2)
Search API	Serper API
Environment	python-dotenv

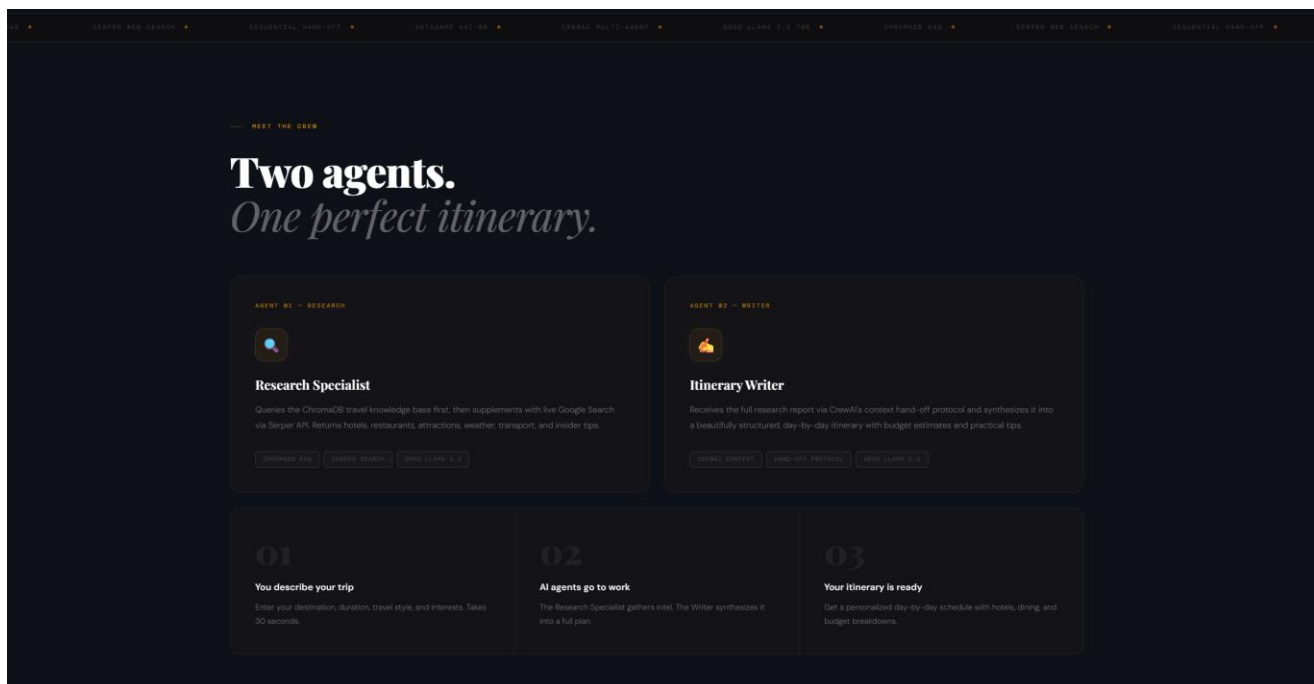
Results & Output

1. Screenshots / Outputs

- Main UI



- Agent Details



- **Travel Preferences**

Tell us how you want to travel.

Your travel preferences

#1 - DESTINATION & DURATION

Tokyo, Paris, Bali, Rajasthan... NUMBER OF DAYS 5

#2 - TRAVEL STYLE

WHO IS TRAVELLING? BUDGET RANGE

Solo Budget (20-23,000/day)

#3 - INTERESTS & ACTIVITIES

WHAT EXCITES YOU?

Food & Dining Culture & History

#4 - SPECIAL REQUIREMENTS (optional)

Vegetarian, halal, wheelchair accessible, avoid spicy food.

+ GENERATE MY ITINERARY

- **Sample (as example)**

Your travel preferences

#1 - DESTINATION & DURATION

Japan main cities like Tokyo, Osaka etc NUMBER OF DAYS 11

#2 - TRAVEL STYLE

WHO IS TRAVELLING? BUDGET RANGE

Solo Luxury (28,000+/day)

#3 - INTERESTS & ACTIVITIES

WHAT EXCITES YOU?

Food & Dining Culture & History Art & Museums Photography Spots Relaxation & Wel... Shopping

#4 - SPECIAL REQUIREMENTS (optional)

Vegetarian, halal, wheelchair accessible, avoid spicy food.

Japan main cities like Tokyo, Osaka etc 11 days Solo Luxury Food & Dining Culture & History Art & Museums Photography Spots Relaxation & Wellness Shopping

+ GENERATE MY ITINERARY

Crafting your 11-day Japan main cities like Tokyo, Osaka etc Itinerary...

TravelCrew

• Output

Your *Japan main cities like Tokyo, Osaka etc* Itinerary

11 DAYS • SOLID • LUXURY • FOOD & DINING • CULTURE & HISTORY • ART & MUSEUMS • PHOTOGRAPHY SPOTS • RELAXATION & WELLNESS • SHOPPING

🌸 🇯🇵 ****Welcome to Japan: A Luxury Travel Itinerary**** 🇯🇵 🌸

=====

Japan, a land of vibrant culture, rich history, and breathtaking beauty, is a destination that will leave you enchanted and inspired. From the neon-lit streets of Tokyo to the historic temples of Osaka, this 11-day luxury travel itinerary is designed to showcase the best of Japan's main cities, tailored to your interests in food, culture, art, photography, relaxation, and shopping.

****Day 1: Arrival in Tokyo**** 🇯🇵

****Morning (9:00 am - 12:00 pm)**:** Arrive at Tokyo's Narita or Haneda airport and take a luxury transfer to your hotel.

****Afternoon (1:00 pm - 5:00 pm)**:** Check-in at the ****Four Seasons Hotel Tokyo at Marunouchi**** 🏨 (approx. \$1,000 per night), a 5-star hotel located in the heart of the city, offering stunning views of the Tokyo Station area. Spend the afternoon exploring the nearby ****Tokyo Imperial Palace**** 🏰, a beautiful park with gardens, moats, and the main palace.

****Evening (6:00 pm - 10:00 pm)**:** Enjoy a welcome dinner at ****Sukiyabashi Jiro**** 🍣 (approx. \$200 per person), a renowned sushi restaurant with a Michelin three-star rating. Must-order dishes include the Omakase (chef's selection) and the highest-quality sushi.

****Day 2: Tokyo**** 🇯🇵

****Morning (9:00 am - 12:00 pm)**:** Visit the ****Tsukiji Outer Market**** 🐟 (free admission), one of the largest fish markets in the world, and try some of the freshest sushi for breakfast at ****Sushi Dai**** 🍣 (approx. \$50 per person).

****Afternoon (1:00 pm - 5:00 pm)**:** Explore the ****Meiji Shrine**** 🏯 (free admission), a serene Shinto shrine located in a tranquil forested area, and take a stroll through the nearby ****Yoyogi Park**** 🌳 (free admission).

****Evening (6:00 pm - 10:00 pm)**:** Enjoy a traditional Japanese dinner at ****Narisawa**** 🍽️ (approx. \$150 per person), a two-Michelin-starred restaurant serving modern Japanese cuisine. Must-order dishes include the Satoyama and the Japanese wagyu beef.

****Day 3: Tokyo**** 🇯🇵

****Morning (9:00 am - 12:00 pm)**:** Visit the ****Ghibli Museum**** 🎬 (approx. \$20 per person), dedicated to the works of Studio Ghibli and its founder Hayao Miyazaki.

****Afternoon (1:00 pm - 5:00 pm)**:** Explore the ****Neon Districts of Shinjuku**** 🌃 (free admission), including the famous ****Kabukicho**** and ****Golden Gai**** areas.

****Evening (6:00 pm - 10:00 pm)**:** Enjoy a drink at the ****New York Bar**** 🍸 (approx. \$20 per person) at the Park Hyatt Tokyo, offering stunning views of the city.

****Day 4: Travel to Osaka**** 🇯🇵

****Morning (9:00 am - 12:00 pm)**:** Take a bullet train (Shinkansen) from Tokyo to Osaka (approx. \$150 per person, one way).

****Afternoon (1:00 pm - 5:00 pm)**:** Check-in at the ****The Ritz-Carlton, Osaka**** 🏨 (approx. \$500 per night), a 5-star hotel located in the heart of the city, offering stunning views of the Osaka Castle area.

****Evening (6:00 pm - 10:00 pm)**:** Enjoy a traditional Osaka dinner at ****Kani Doraku**** 🍣 (approx. \$50 per person), a famous crab restaurant with a unique and delicious menu.

****Day 5: Osaka**** 🇯🇵

****Morning (9:00 am - 12:00 pm)**:** Visit the ****Osaka Castle**** 🏰 (approx. \$10 per person), a majestic castle that played a crucial role in Japan's unification.

****Afternoon (1:00 pm - 5:00 pm)**:** Explore the ****Dotonbori**** 🌃 (free admission), a popular entertainment district known for its colorful neon lights, street food, and unique attractions.

****Evening (6:00 pm - 10:00 pm)**:** Enjoy a dinner at ****Tsurutontan**** 🍜 (approx. \$30 per person), a famous udon noodle restaurant with a wide variety of delicious dishes.

****Day 6: Osaka**** 🇯🇵

****Morning (9:00 am - 12:00 pm)**:** Visit the ****Shinsaibashi-suji Shopping Arcade**** 🛍️ (free admission), one of the longest shopping streets in Japan, offering a wide range of products and souvenirs.

****Afternoon (1:00 pm - 5:00 pm)**:** Explore the ****Umeda Sky Building**** 🏙️ (approx. \$20 per person), offering stunning views of the city from its rooftop observatory.

****Evening (6:00 pm - 10:00 pm)**:** Enjoy a drink at the ****Liberation**** 🍹 (approx. \$15 per person), a unique bar with a wide selection of craft beers and cocktails.

****Day 7: Kyoto**** 🇯🇵

****Morning (9:00 am - 12:00 pm)**:** Take a train from Osaka to Kyoto (approx. \$10 per person, one way).

****Afternoon (1:00 pm - 5:00 pm)**:** Visit the ****Fushimi Inari Shrine**** 🏯 (free admission), famous for its thousands of vermilion torii gates that form a tunnel up the mountain.

****Evening (6:00 pm - 10:00 pm)**:** Enjoy a traditional Kyoto dinner at ****Gion Nanba**** 🍽️ (approx. \$100 per person), a renowned restaurant serving high-quality kaiseki cuisine.

****Day 8: Kyoto**** 🇯🇵

****Morning (9:00 am - 12:00 pm)**:** Visit the ****Arashiyama Bamboo Grove**** 🎋 (free admission), a serene and picturesque area with towering bamboo stalks.

****Afternoon (1:00 pm - 5:00 pm)**:** Explore the ****Kinkaku-ji Temple**** 🏯 (approx. \$5 per person), a beautiful temple with a stunning golden pavilion.

****Evening (6:00 pm - 10:00 pm)**:** Enjoy a traditional Japanese tea ceremony at ****Camellia**** 🍵 (approx. \$20 per person), a unique and peaceful experience.

Day 9: Return to Tokyo 🇯🇵

- * **Morning (9:00 am - 12:00 pm)**: Take a train from Kyoto to Tokyo (approx. \$150 per person, one way).
- * **Afternoon (1:00 pm - 5:00 pm)**: Spend the afternoon at leisure, exploring the city or relaxing at your hotel.
- * **Evening (6:00 pm - 10:00 pm)**: Enjoy a farewell dinner at **Nihonryori Ryugin** 🍽️ (approx. \$200 per person), a Michelin three-starred restaurant serving modern Japanese cuisine.

Day 10: Tokyo 🇯🇵

- * **Morning (9:00 am - 12:00 pm)**: Visit the **Asakusa** 🗼️ (free admission), a historic district with a famous Senso-ji Temple and Nakamise shopping street.
- * **Afternoon (1:00 pm - 5:00 pm)**: Explore the **Akihabara** 📺 (free admission), a district known for its electronic shops, anime, and manga culture.
- * **Evening (6:00 pm - 10:00 pm)**: Enjoy a drink at the **Bar High Five** 🍹 (approx. \$20 per person), a unique bar with a wide selection of craft cocktails.

Day 11: Departure ✈️

- * **Morning (9:00 am - 12:00 pm)**: Spend the morning at leisure, shopping for last-minute souvenirs or exploring the city.
- * **Afternoon (1:00 pm - 5:00 pm)**: Check-out of your hotel and take a luxury transfer to the airport for your departure flight.

Hotel Recommendations 🏨

- * **Four Seasons Hotel Tokyo at Marunouchi** (approx. \$1,000 per night)
- * **The Ritz-Carlton, Osaka** (approx. \$500 per night)
- * **Park Hyatt Tokyo** (approx. \$600 per night)

Restaurant Suggestions 🍽️

- * **Sukiyabashi Jiro** (approx. \$200 per person)
- * **Narisawa** (approx. \$150 per person)
- * **Kani Doraku** (approx. \$50 per person)
- * **Tsunotontan** (approx. \$30 per person)
- * **Gion Namba** (approx. \$100 per person)

Practical Tips 📌

- * Japan has a unique culture and customs, so it's essential to respect local traditions and rules.
- * Learn some basic Japanese phrases, such as "onnichiwa" (hello), "arigatou" (thank you), and "sumimasen" (excuse me).
- * Try to avoid traveling during peak hours (7:00 am - 9:00 am and 5:00 pm - 7:00 pm) to avoid crowds and congestion.
- * Don't tip at restaurants or bars, as it's not customary in Japan.
- * Carry a portable Wi-Fi hotspot or purchase a prepaid SIM card for data and phone access.

Estimated Daily Budget Breakdown 💰

- * Accommodation: \$500 - \$1,000 per night
- * Food and drink: \$100 - \$300 per day
- * Transportation: \$100 - \$200 per day
- * Attractions and activities: \$50 - \$100 per day
- * Total: \$750 - \$1,600 per day

Note: The estimated daily budget breakdown is approximate and can vary depending on individual preferences and exchange rates.

2. Reports / Models

- Research Report (generated by Researcher Agent)

```
Task Completed

Name: Research the following travel request thoroughly: 11 days in Japan main cities like Tokyo, Osaka etc for solo. Budget: Luxury.
Interests: Food & Dining, Culture & History, Art & Museums, Photography Spots, Relaxation & Wellness, Shopping. Provide hotels,
restaurants, attractions, day-by-day schedule, transport tips, and budget breakdown.

You MUST gather and include:
1. Top 3-5 hotels with names, price range, and key features
2. Top 5 restaurants with cuisine type and must-try dishes
3. Top 8-10 local attractions and activities with brief descriptions
4. Best time to visit and weather information
5. Local transport options
6. Practical travel tips (currency, language, customs, safety)

Be specific with names, locations, and details. No vague suggestions.
Agent: Hospitality Research Specialist
```

```
Agent Started

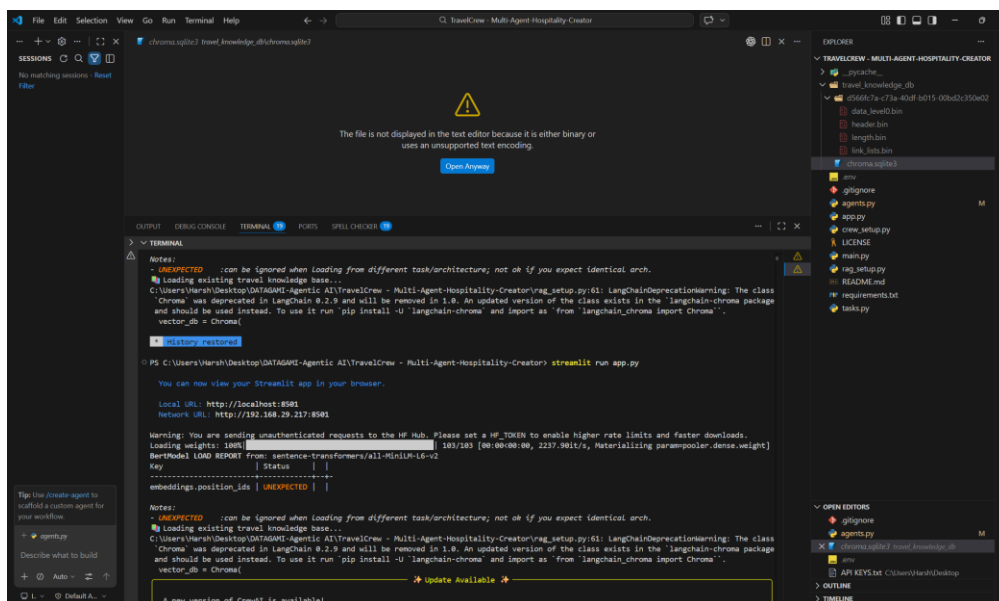
Agent: Travel Itinerary Writer

Task: Using the research provided, create a complete day-by-day travel itinerary for: 11 days in Japan main cities like Tokyo, Osaka etc
for solo. Budget: Luxury. Interests: Food & Dining, Culture & History, Art & Museums, Photography Spots, Relaxation & Wellness, Shopping.
Provide hotels, restaurants, attractions, day-by-day schedule, transport tips, and budget breakdown.

The itinerary MUST include:
1. A warm, engaging introduction to the destination
2. Day-by-day schedule (morning, afternoon, evening) with timings
3. Hotel recommendations with why each is a good choice
4. Restaurant suggestions per meal with must-order dishes
5. Attractions and activities woven into the daily schedule
6. Practical tips section at the end
7. Estimated daily budget breakdown

Format it beautifully with clear sections, emojis for readability, and a friendly but professional tone.
```

- Vector embeddings stored in ChromaDB



```
File Edit Selection View Go Run Terminal Help
chroma.sqlite3 travel_knowledge_db\chroma.sqlite3

No file is displayed in the text editor because it is either binary or
uses an unsupported text encoding.
Open Anyway

OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER
- UNEXPECTED - can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- Loading existing travel knowledge base...
C:\Users\Varsh\Desktop\DATAGAMI-Agentic AI\TravelCrew - Multi-Agent-Hospitality-Creator\vrag_setup.py:61: LangChainDeprecationWarning: The class
'Chroma' was deprecated in LangChain 0.2.9 and will be removed in 1.0. An updated version of the class exists in the 'langchain-chroma' package
and should be used instead. To use it run 'pip install -U 'langchain-chroma' and import as 'from langchain_chroma import Chroma'.
vector_db = Chroma()

- History restored

PS C:\Users\Varsh\Desktop\DATAGAMI-Agentic AI\TravelCrew - Multi-Agent-Hospitality-Creator> streamlit run app.py

You can now view your Streamlit app in your browser.
Local URL: http://localhost:8501
Network URL: http://192.168.29.217:8501

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100% | 160/168 [46:00:00.00, 223.8Mi/s, Materializing parampooler.dense.weight]
BERTModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key | Status | |
-----|-----|
embeddings.position_ids | UNEXPECTED | |

Notes:
- UNEXPECTED - can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- Loading existing travel knowledge base...
C:\Users\Varsh\Desktop\DATAGAMI-Agentic AI\TravelCrew - Multi-Agent-Hospitality-Creator\vrag_setup.py:61: LangChainDeprecationWarning: The class
'Chroma' was deprecated in LangChain 0.2.9 and will be removed in 1.0. An updated version of the class exists in the 'langchain-chroma' package
and should be used instead. To use it run 'pip install -U 'langchain-chroma' and import as 'from langchain_chroma import Chroma'.
vector_db = Chroma()

A new version of CrawlAI is available!

EXPLORER
TRAVELCREW - MULTI-AGENT-HOSPITALITY-CREATOR
- __pycache__
- travel_knowledge_db
  - data_loader.py
  - header.py
  - length.py
  - link.py
  - chroma.sqlite3
- vrag
  - __pycache__
  - agents.py
  - app.py
  - crew_setup.py
  - LICENSE
  - main.py
  - rag_setup.py
  - README.md
  - requirements.txt
  - tasks.py

OPEN EDITORS
- chroma.sqlite3 travel_knowledge_db
- agents.py
- app.py
- rag_setup.py
- API_KEYS.txt C:\Users\Varsh\Desktop\

OUTLINE
TIMELINE
```

3. Key Outcomes

- Successfully developed a fully functional multi-agent AI system for automated travel planning
- Significantly reduced manual effort in itinerary creation through intelligent automation
- Generated high-quality, structured, and personalized travel itineraries
- Achieved efficient data retrieval using a RAG-first approach with ChromaDB
- Ensured fast response times through Groq LLM inference and local vector database access

Conclusion

TravelCrew successfully demonstrates the implementation of an Agentic AI system using a multi-agent architecture. By integrating Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), and real-time web search, the system delivers accurate, personalized, and structured travel itineraries.

The project highlights key concepts such as AI system design, multi-agent orchestration, and efficient data flow within a modular architecture. Its scalable and extensible design makes it suitable as a foundation for real-world intelligent travel planning systems.

Future Scope & Enhancements

- Extend the system by introducing additional specialized agents such as:
 - Budget Optimization Agent
 - Weather Prediction Agent
 - Flight Booking Agent
- Integrate real-time booking and reservation APIs for end-to-end travel execution
- Deploy the system on cloud platforms such as AWS or Google Cloud for scalability
- Implement user authentication and itinerary history management
- Enhance the user interface with interactive maps and visualizations
- Develop a mobile application for improved accessibility and user experience